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标题	Recent Advances in Ruthenium-Catalyzed Hydrogenation Reactions of Renewable Biomass-Derived Levulinic Acid in Aqueous Media
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摘要（原文节选）

Levulinic acid (LA) is classified as a key platform chemical for the development of future biorefineries, owing to its broad spectrum of potential applications and because it is simply available from lignocellulosic biomass through inexpensive and high-yield production routes. Catalytic hydrogenation reactions of LA into the pivotal intermediate compound γ-valerolactone (GVL), and beyond GVL to yield valeric acid (VA), 1,4-pentanediol (1,4-PDO), and 2-methyltetrahydrofuran (2-MTHF) have gained considerable attentio ⋯

数据表 · Table 1（26 行，含数值行 26 行）

Ruthenium-based heterogeneous catalysts on solid supports and water-dispersible catalytic nanoparticles for the hydrogenation of LA toward GVL.

Catalysts	Molar ratio LA/M	T (°C)	PH2 (bar)	t (h)	Solvent	Conversion LA (mol%)	Selectivity GVL (mol%)	TOFa (h−1)
0.5% Ru/Al2O3	8,707	130	40	3	Water	13.8	99.9	401
0.5% Ru/SiO2	8,707	130	40	3	Water	80.1	99.8	2,325
0.5% Ru/ZrO2	8,707	130	40	3	Water	80.3	99.9	2,331
0.5% Ru/TiO2	8,707	130	40	3	Water	95.4	99.9	2,769
1% Ru/TiO2	4,353	70	40	0.25	Water	44	99.9	7,662
1% Ru/TiO2	4,353	130	40	0.5	Water	100	99.9	8,707
1% Ru/TiO2	4,353	130	40	1	Ethanol	74.9	47.7	3,261
1% Ru/TiO2	4,353	130	40	1	1,4-Dioxane	30.7	99.9	1,337
1% Ru/OMS	3,831	100	30	1	Water	99.9	99.8	3,420
RuCl3 · 3H2O/TiO2	4,350	90	45	4	Water	86	92	1,152
RuNO(NO3)3/TiO2	4,350	90	45	4	Water	77	94	824
5% Ru/ZrO2	2,100	170	27	7	Water	99.0	99.9	297
2% Ru/FLG	1,460	20	40	8	Water	99.3	97.7	184
5% Ru/C	359	130	12	2.7	Water	99.5	86.6	133
0.64% Ru/TiO2	247	150	35	5	Water	100	93	49
5% Ru/TiO2	106	150	35	5	Water	100	90	21
0.83% Ru/TiO2 ultrathin	4,878	100	40	1.5	Water	100	99.1	19,045b
Ru/NCS	10,000	70	40	1	Water	51	100	9,858
Ru/CS	10,000	70	40	1	Water	32	100	6,985
Ru/TiO2	505	150	20	5	Water	100	94.8c	101
RuCl3 · 3H2O/PEG400	40	130	20	1	PEG/Water	99	99	40
Ru3(CO)12	1,720	130	5	12	Water	100	100	143
RuNHC	1,000	130	12	2.7	Water	99	96	361
RuNHC	1,000	130	12	2.7	THF	1	1	3
Ru@PEG-CD	1,300	80	40	4	Water	97	99	315
Ru@PVP	1,300	80	40	4	Water	93	99	302

脚注（逐字）：aDefined as mole of hydrogenated LA per mole of ruthenium per hour.bDefined as mole of hydrogenated LA per number of surface ruthenium atoms per hour.cSelectivity toward 2-MTHF: 5.2 mol%.

正文对应句（逐字）：which might be due to an H/D exchange between D2O and GVL. Thus, is proven that the promotional effect of the aqueous solvent in the LA hydrogenation is based on the involvement of water after dissociation as a reactant in the mechanism of this aqueous-phase catalytic reaction. Michel et al. (2014)

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